

1.	Write Regular Expressions for the following: a. To accept strings ending with '.' or '\ ' and the length of the string between 4 to 7. b. To accept colour and color.	[2]
2.	Consider the following corpus to implement the Kneser-Ney smoothing algorithm for the trigram model, "Natural language processing is essentially the study of the meaning of the words a human says or writes. Natural language processing is all around us all the time. Natural language processing is all about mimicking our own language patterns. Natural language processing can also improve the efficiency of business transactions. Natural language processing is the application of computer technology." Assuming $d=0.75$, it is also given that no of unigrams = 78, no of bigrams = 122, no of trigrams = 130. What will be the value of $P_{kn}(\text{processing} \text{language})$?	[2]
3.	For the corpus given in Q2, if the threshold $k=1$, Find the value of $P(\text{is} \text{processing})$ for Katz Backoff model for the bigram model.	[2]
4.	Consider the following sample corpus: <s> I am grut </s> <s> grut I am </s> <s> I do not like green leafs </s> Learn a bigram language model using the corpus and find the probability of the sentence: <s> I am grut green </s>	[2]
5.	Using the corpus data given in Q4, calculate the bigram language model with add-K smoothing and find the probability for the same sentence where $K=0.2$.	[2]
6.	Draw an FST which correctly recognizes the affix -ly associated with regular adverbs in English.	[2]
7.	Draw a DFA that accepts tokens representing English word tokens that may contain at most one apostrophe, and the apostrophe cannot occur as the first or last character	[2]
8.	Consider the corpus given in Q4, Find out total number of unique trigrams.	[2]
9.	Consider a small corpus: I grut am I grut eat I leaf and am do not fruits. Find the probability of seeing the word "is" using good turing smoothing for the unigram model.	[2]
10.	Take the corpus in Q2 and find the TF-IDF vector for the sequence using only unigrams: Natural Language Processing	[2]